MINDWALLET

Cognitive Design & Interaction Implementation Report

Mental Models • Execution & Evaluation • Distributed Cognition
External/Extended Cognition • Embodied Interaction

Course Name: User Interface and User Experience Design

Course Code: ISWE407L

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Personal Finance Platform

1. Project Overview

MindWallet is a human-centered personal finance platform designed to make financial management easier to understand, operate and act upon. The implementation combines transaction tracking, budgeting, savings planning, financial health monitoring, subscription management, analytics, behavioral insights, purchase simulation and reporting within a single interface.

The interaction design is structured around familiar financial concepts and visible feedback. Rather than requiring users to remember financial information or perform repeated calculations themselves, MindWallet stores information, processes it across its modules and presents the results through scores, progress indicators, charts, alerts, insights and reports.

1.1 Main areas considered

- Dashboard and financial overview
- Income and expense tracking
- Budget planning
- Savings goals
- Financial Health Score
- Subscription management
- Spending analytics
- Smart Insights
- Purchase Simulator
- Monthly Reports
- Settings, user modes and accessibility controls

2. Mental Model

MindWallet uses familiar financial concepts to create a predictable interaction model. A user can approach the application with an existing understanding of income, expenses, budgets, savings, subscriptions and financial goals, and map those concepts directly onto the interface.

Incorporation of Mental Models (5 Core Models)

A mental model represents an individual's thought process about how something works in the real world. In **MindWallet**, the interface aligns digital financial interactions with familiar real-world concepts such as a wallet, spending limits, saving toward a goal, recurring commitments, and making purchase decisions. These models reduce cognitive friction by allowing users to understand financial information through concepts they already recognize.

Model 1: The Physical Wallet / Money Overview Model

- **Real-World Concept:** A physical wallet gives a person an immediate sense of how much money they have available and what financial resources they are carrying.
- **MindWallet UI Implementation:** The **Dashboard** acts as a digital financial wallet. It presents **Total Balance**, **Income**, **Expenses**, and **Saved This Month** as large summary cards. The dashboard also combines the Financial Health Score, budget status, savings goals and recent transactions, allowing users to understand their overall financial position from one place.

Model 2: The Household Budget / Spending Limit Model

- **Real-World Concept:** People commonly manage household spending by assigning a fixed amount of money to categories such as groceries, transport, dining and entertainment. Once the allocated amount is nearly used, they know they need to control further spending.
- **MindWallet UI Implementation:** The **Budget Planner** represents each spending category as a defined financial limit. Categories such as **Groceries, Dining Out, Transport and Entertainment** display the amount spent against the allocated budget using visual progress bars. For example, Groceries can be shown as **₹6,200 / ₹8,000**, while Dining Out can show **₹5,850 / ₹5,000**, immediately communicating whether spending is within the planned limit.
- Budget conditions are represented using **Healthy, Warning and Critical** states, allowing users to recognize increasing spending pressure without performing the comparison mentally.

Model 3: The Piggy Bank / Savings Goal Model

- **Real-World Concept:** A piggy bank represents gradually putting money aside until enough has been accumulated for a desired purpose. The physical amount saved can be understood as progress toward a target.
- **MindWallet UI Implementation:** The **Savings Goals** module uses a digital piggy-bank mental model. Users create specific financial goals and monitor their progress toward a target amount. Circular progress rings visually communicate how much of the goal has already been achieved.
- For example, an **Emergency Fund** can display **₹75,000 of ₹1,00,000 · 75%**, allowing the user to immediately understand both the current amount and the percentage completed.
- Users can also add deposits and view estimated completion information, connecting regular saving behaviour with the eventual goal.

Model 4: The Recurring Bill / Subscription Commitment Model

- **Real-World Concept:** Bills and memberships such as streaming services, gym memberships or subscriptions create recurring financial commitments. Unlike one-time purchases, they continue to consume money periodically until they are cancelled.
- **MindWallet UI Implementation:** The **Subscription Manager** represents subscriptions as recurring financial commitments. It tracks **monthly and annual recurring costs**, billing information and the ongoing financial burden created by subscriptions.
- The module also provides a **cancellation savings simulator**, allowing users to understand how much money could be saved by cancelling a recurring service. This transforms an abstract recurring expense into a concrete financial consequence.

Model 5: The "Can I Afford This?" Purchase Decision Model

- **Real-World Concept:** Before making an expensive purchase, people commonly ask themselves, "Can I afford this?" They mentally consider the purchase price, the money they will have left afterward, and whether the purchase will interfere with other financial goals.

- **MindWallet UI Implementation:** The **Purchase Simulator** directly models this real-world decision process through the **"Can I Afford This?"** interface. Instead of requiring the user to perform multiple calculations mentally, MindWallet calculates the **price impact, net balance after purchase and savings-goal delay**.
- The system then converts the calculation into a simple recommendation:

  Proceed Safe

  Wait 30 Days

  Avoid Purchase

- This allows the user to treat a potential purchase as a clear financial decision rather than simply another transaction.

2.1 How the mental model is implemented

Interface element	Mental model established	Implementation
Dashboard	A financial control centre	Presents balance, expenses, savings, financial health, budgets and recent activity together.
Tracker	A record of money coming in and going out	Provides transaction entry, categories, amount, date and transaction type.
Budget Planner	A spending limit	Shows category spending against defined monthly limits using progress bars and status levels.
Savings Goals	Saving toward a specific target	Uses goal names, target amounts, monthly deposits and progress indicators.
Financial Health Score	A simple measure of financial condition	Combines financial factors into a 0–100 score with an explanatory message and tags.
Subscriptions	Recurring financial commitments	Stores service, recurring amount, billing cycle, category and renewal date.
Purchase Simulator	Checking affordability before buying	Uses the question "Can I Afford This?" and returns financial consequences and a recommendation.
Reports	A monthly financial statement	Provides a clean, printable representation of financial activity.

2.2 Recognition over recall

The interface keeps important options visible instead of requiring users to remember where functions are located. The navigation exposes the major financial tasks directly, while forms use descriptive field labels and action names. For example, transaction entry is presented as "Log New Transaction" with fields for Merchant / Source Name, Transaction Type, Amount, Category and Date, followed by "Save Transaction".

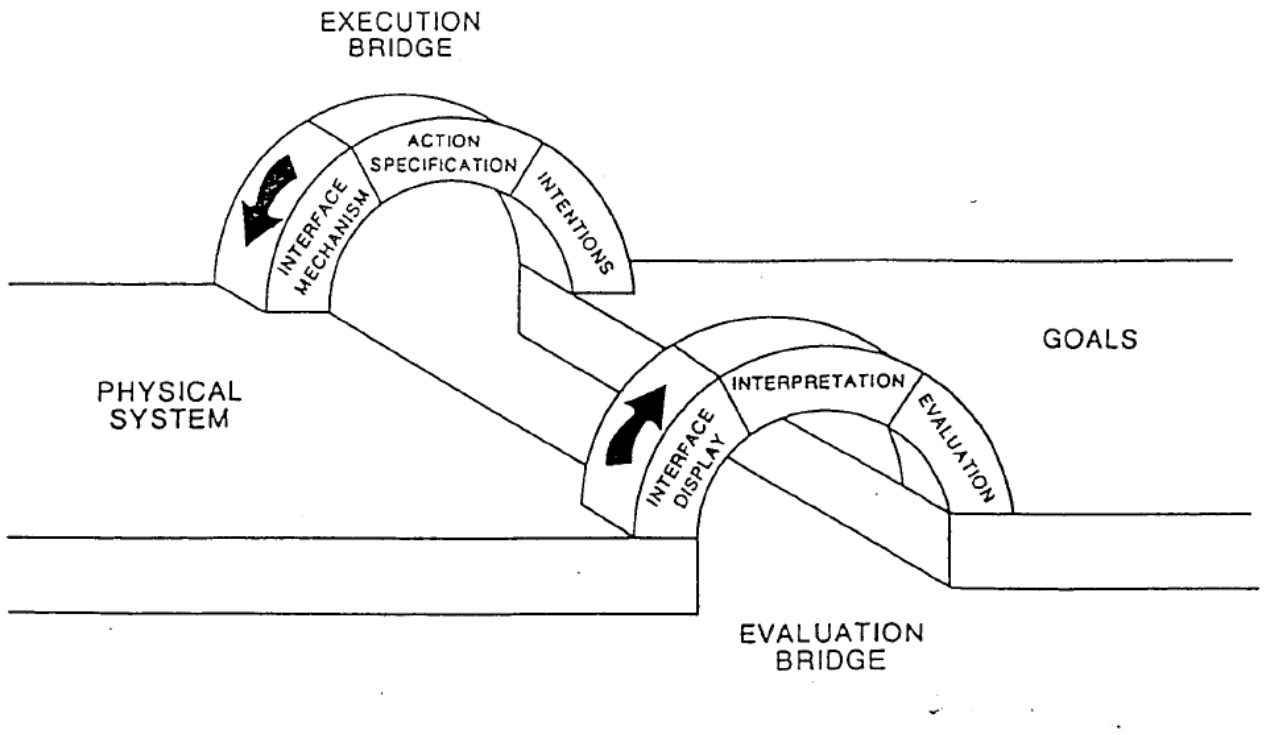
This structure allows users to recognize the required action from the interface itself. The same approach is used for budget limits, savings goals and subscriptions.

2.3 Accessibility-supported mental models

MindWallet provides Standard, Accessible, Senior and Simplified modes. These modes preserve the same underlying financial concepts while changing presentation and interaction density. Senior Mode increases text size and touch targets, while Simplified Mode hides non-essential secondary details. The accessibility panel also provides high contrast, reduced motion and text scaling controls.

3. Bridge of Execution

The bridge of execution describes how effectively a user's financial intention can be translated into actions on the MindWallet interface. The implementation reduces this gap through visible navigation, clearly labelled controls, structured forms and direct actions.



3.1 General execution path

USER GOAL — What does the user want to accomplish?

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INTENTION — Choose the appropriate MindWallet function

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ACTION — Navigate, select, enter information or activate a control

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SYSTEM PROCESSING — MindWallet stores or calculates the requested information

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RESULT — The requested financial state or decision information is produced

3.2 Execution implementation by task

User goal	Execution path in MindWallet	Why the path is clear
Record an expense	Tracker → Log New Transaction → enter merchant/type/amount/category/date → Save Transaction	The task and final action are explicitly named.
Set a category budget	Budget Planner → choose category → enter monthly limit → Update Budget Threshold	The form directly represents the intended budgeting task.
Create a savings goal	Savings Goals → New Goal → enter name/target/monthly deposit/category → Save Goal	The required information follows the user's real-world goal-setting process.
Track a subscription	Subscriptions → add subscription → service/amount/cycle/category/renewal date → Track Subscription	Recurring financial commitments are handled in one dedicated task flow.
Evaluate a purchase	Simulator → enter purchase information → calculate affordability → review recommendation	The function is framed around the user's decision rather than a technical calculation.
Review finances	Dashboard → view balance, health, budget, savings and recent transactions	Important information is available from a single starting point.

4. Bridge of Evaluation and Visual Feedback

The bridge of evaluation describes how effectively MindWallet communicates the result of an action back to the user. The implementation uses visual representations and concise textual feedback so users can understand current financial state without reconstructing it from raw data.

4.1 Evaluation path

ACTION — User completes a financial task

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SYSTEM RESPONSE — Data is updated or a calculation is performed

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VISUALIZATION — Updated values, progress, charts, scores or messages are displayed

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INTERPRETATION — User understands the meaning of the result

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EVALUATION — User decides whether the financial goal or task has been achieved

4.2 Visualization used for evaluation

Visualization / feedback	What the user can understand
Financial Health Score meter	Overall financial condition through a 0–100 score and supporting explanation.
Budget progress bars	Current category spending relative to the allowed limit.
Savings progress rings	Progress toward a savings target.
Spending analytics	Distribution and comparison of spending across categories.
Budget status indicators	Whether spending is within a healthy range, requires attention or has exceeded the limit.
Smart Insights	A plain-language interpretation of financial behavior.
Simulator recommendation	The practical consequence of a purchase and an actionable recommendation.
Notifications	Recent financial alerts such as budget notices and

	recurring charges.
Monthly Reports	A consolidated view of financial activity for review.

Evaluation principle applied in MindWallet

The system does not stop after accepting an input. It presents the resulting financial state in a form that can be seen, compared and interpreted by the user.

5. Complete Interaction Flows

5.1 Transaction recording flow

Goal: Record a financial transaction

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Tracker: Open the transaction function

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Input: Merchant, transaction type, amount, category and date

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Action: Save Transaction

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Feedback: Updated transaction information and financial summaries

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Evaluation: Confirm that the record and financial state are correct

5.2 Budget management flow

Goal: Control spending in a category

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Budget Planner: Select a category and define its monthly limit

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Action: Update Budget Threshold

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Feedback: Category spending is represented against the limit

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Visualization: Progress bar and status communicate current position

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Evaluation: User determines whether spending is under control

5.3 Savings goal flow

Goal: Save toward a defined target

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Savings Goals: Create a goal

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Input: Goal name, target amount, monthly deposit and category

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Action: Save Goal

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Feedback: Goal progress and completion information

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Evaluation: User assesses progress toward the target

5.4 Purchase decision flow

Goal: Decide whether to make a purchase

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Simulator: Enter purchase information

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Processing: Calculate price impact, remaining balance and savings-goal delay

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Visualization: Show consequences and recommendation

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Decision: Proceed Safe / Wait 30 Days / Avoid Purchase

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Evaluation: User decides whether the purchase fits the financial situation

6. Distributed Cognition

MindWallet distributes financial reasoning across the user and the system. The user provides goals, inputs and judgements, while the application stores information, performs calculations, transforms records and presents the results through specialized modules.

6.1 Distribution of cognitive work

Component	Role in MindWallet
User	Defines financial goals, enters information, interprets results and makes decisions.
Central data store	Maintains financial records so the user does not need to remember every transaction.
Transaction module	Organizes financial events into structured records.
Budget module	Transforms transaction information into category-level spending status.
Savings module	Transforms targets and deposits into progress information.
Health module	Combines financial factors into a single diagnostic representation.
Analytics module	Transforms financial records into visual spending information.
Insights module	Transforms financial patterns into understandable behavioural guidance.
Simulator	Performs financial consequence calculations for prospective purchases.
Reports	Transforms accumulated financial information into a reviewable statement.

6.2 Information movement

User provides financial information

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MindWallet stores the information

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Relevant modules process the information

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Results are transformed into visual and textual representations

↓

User interprets the representations

↓

User makes the next financial decision

7. External / Extended Cognition

MindWallet acts as an external support system for remembering, organizing, comparing and calculating financial information. Important information is kept visible in the interface, while calculations are performed by the application rather than being left entirely to the user.

7.1 Memory support

MindWallet feature	Externalized information
Transaction history	Past financial events, merchants, categories, dates and amounts.
Budget Planner	Category limits and current spending position.
Savings Goals	Target amount, monthly deposit and progress.
Subscription Manager	Recurring amounts, billing cycles and renewal dates.
Notifications	Recent budget and recurring-payment reminders.
Monthly Reports	Consolidated historical financial information.

7.2 Computational offloading

Several MindWallet features move calculation work from the user's mind to the application. The Financial Health Score combines multiple financial indicators into a 0–100 score. The Purchase Simulator calculates purchase impact, net balance after purchase and possible savings-goal delay. Savings Goals provide estimated completion information. These calculations are then presented in forms that are easier to interpret.

7.3 Visual externalization

The Dashboard uses visual meters, progress bars, score displays and summary cards to make financial information available at a glance. This is particularly important for information that would otherwise require comparing several numbers mentally.

8. Embodied Interaction

MindWallet supports direct interaction in which users physically manipulate interface controls to carry out financial tasks. The interaction is action-oriented: users navigate, click or tap controls, enter values, select categories and activate commands that directly correspond to financial operations.

8.1 Where embodied interaction appears

Interaction	Implementation in MindWallet
Navigation	Users click or tap Dashboard, Tracker, Budget Planner, Savings Goals and other modules.
Data entry	Users physically enter merchant names, amounts, dates and goal information.
Selection	Users select transaction types, financial categories, billing cycles and goal categories.
Action controls	Buttons such as Save Transaction, Save Goal and Track Subscription directly trigger tasks.
Responsive navigation	The interface provides a menu control for navigation on smaller screens.
Accessibility controls	Users directly activate high contrast, reduced

	motion and text scaling controls.
Text-size controls	A−, Reset and A+ controls allow direct adjustment of text presentation.

8.2 Scope of embodied interaction

The implementation demonstrates embodied interaction through direct physical interaction with standard interface controls and responsive layouts. It does not rely on specialized gestures such as pinch-to-zoom or motion sensing; therefore, the report limits the claim to the interaction mechanisms actually present in MindWallet.

9. Integrated Cognitive Design Evaluation

All concepts work together across the MindWallet experience. The mental model makes financial functions recognizable; the execution bridge makes those functions actionable; the evaluation bridge makes system results understandable; distributed cognition shares the reasoning process between the user and the application; external/extended cognition reduces memory and calculation demands; and embodied interaction provides direct control over the system.

Design objective	MindWallet implementation	Expected benefit
Make the system understandable	Familiar financial concepts and descriptive labels	Users can quickly recognize functions and expected outcomes.
Make actions easy to perform	Dedicated navigation and structured forms	Lower uncertainty during task execution.
Make results visible	Scores, bars, rings, charts, alerts and reports	Users can interpret financial state without reconstructing it mentally.
Reduce memory demand	Persistent transactions, budgets, goals and subscriptions	Users can rely on the system as an external record.
Reduce calculation demand	Health Score, goal calculations and Purchase Simulator	Complex financial reasoning is partially offloaded to the application.
Support different users	Standard, Accessible, Senior and Simplified modes	Presentation and interaction can be adapted to user needs.
Support decision-making	Smart Insights and Purchase Simulator	Users receive actionable information instead of only raw financial data.

9.1 Overall assessment

- MindWallet has a clear and consistent financial interaction model.
- The major tasks are represented as recognizable user goals rather than technical operations.
- The interface provides direct paths from intention to action through dedicated modules and clearly named controls.
- Visual feedback is central to the evaluation of financial state, particularly through progress indicators, scores and charts.
- The application reduces dependence on internal memory by maintaining financial records and recurring information.
- The application reduces mental calculation effort through automated financial analysis and simulation.
- Accessibility settings strengthen the cognitive design by reducing visual complexity and allowing presentation changes.
- The strongest cognitive-support features are the Dashboard, Budget Planner, Savings Goals, Financial Health Score and Purchase Simulator.

10. Research UX (User Experience Research)

MindWallet's User Experience Research focuses on understanding how users manage personal finances, identify difficulties in tracking expenses and budgets, and determine how the interface can support clearer and more confident financial decisions. The research particularly considers **cognitive load, financial literacy, accessibility, and decision support**.

Heuristic Evaluation & Usability Insights

System Status Visibility (Nielsen Heuristic #1)

MindWallet keeps users informed about their financial status by presenting important information directly within the interface. Users can understand their current financial condition without having to navigate through multiple screens. Financial Health Score provides an overall indication of the user's financial condition.Budget progress indicators show the amount spent compared with the assigned category limit. Savings progress visually communicates how close the user is to achieving a savings goal.Budget categories use **HEALTHY, WARNING, and CRITICAL** states to communicate spending conditions clearly.

Error Prevention over Error Recovery (Nielsen Heuristic #5)

Financial applications should help users identify potential problems before they make decisions that negatively affect their finances. MindWallet incorporates preventive decision support rather than relying only on correcting mistakes afterward.Budget limits make users aware when their spending is approaching or exceeding a planned amount.Smart Insights provide contextual guidance based on spending behaviour.The **Purchase Simulator** allows users to evaluate the financial effect of a purchase before making it.The simulator provides recommendations such as **Proceed Safe, Wait 30 Days, and Avoid Purchase**, helping users reconsider unnecessary or risky purchases.

Recognition Rather than Recall (Nielsen Heuristic #6)

MindWallet reduces the need for users to remember financial information by keeping important information visible and using familiar visual representations.The dashboard presents balance, income, expenses, savings, and other important financial information together.Spending categories are clearly labelled so users can identify where their money is going.Savings goals use visual progress indicators to communicate achievement.Recent transactions allow users to quickly recognize their latest financial activities.

Accessibility Constraints (WCAG 2.2 AA Target)

Adaptive User Modes :

MindWallet provides multiple interface modes to accommodate different user requirements. These include **Standard Mode, Accessible Mode, Senior Mode, and Simplified Mode**.

Text Scaling :

The Accessibility Control Panel provides text scaling controls that allow users to increase or decrease the interface text size. This supports users who require larger typography for improved readability.

High Contrast and Reduced Motion :

MindWallet includes **High Contrast** and **Reduced Motion** controls. These options help users who require improved visual distinction or prefer reduced animation and movement.

Reduced Cognitive Load :

Simplified Mode uses **progressive disclosure** to hide non-essential secondary information. This prevents users from being overwhelmed by too much financial information at once and directly addresses the research finding that excessive information can increase cognitive load.

11. Research for User (User Personas & Behavioral Profiles)

Persona 1: The Beginner Money Manager (Primary User)

Name: Arjun Kumar
Age: 21 years old
Occupation: Student
Location: Hyderabad
Living Situation: Lives with family and is beginning to manage his personal monthly expenses.

Physical & Digital Characteristics:

1.Arjun is comfortable using smartphones and digital payment applications but has limited knowledge of financial planning and budgeting.

2.He prefers simple interfaces and quick access to information rather than detailed financial analysis.

3.He regularly makes small day-to-day purchases and may not consistently record every expense.

Core Frustrations:

Difficulty understanding where his money is being spent.

Forgetting to record small daily expenses.

Difficulty understanding financial terminology and complex charts.

Uncertainty about whether an expensive purchase is financially affordable.

Design Requirements:

A simple dashboard displaying important financial information immediately.

Quick Add Expense for faster transaction recording.

Clear visual budget progress indicators.

Plain-language Smart Insights.

Purchase Simulator for evaluating expensive purchases.

Persona 2: The Budget-Conscious Working Professional (Secondary User)

Name: Priya Sharma
Age: 30 years old
Occupation: IT Professional
Location: Bengaluru

Living Situation: Lives independently and manages monthly income, expenses, subscriptions, and savings.

Physical & Digital Characteristics:

Priya is comfortable using digital financial applications and regularly manages multiple financial categories.

She has recurring subscriptions and monthly financial commitments that need to be monitored.

She prefers quickly reviewing her financial condition instead of manually analysing every transaction.

Core Frustrations:

Difficulty maintaining monthly spending limits consistently.

Recurring subscriptions increasing monthly expenses without being noticed.

Difficulty balancing regular spending with long-term savings.

Limited visibility into how spending behaviour affects overall financial health.

Design Requirements:

Budget Planner with category-level spending limits.

Subscription Manager for monitoring recurring expenses.

Spending Analytics for understanding spending patterns.

Savings Goals with visual progress tracking.

Financial Health Score for a quick overall financial assessment.

Persona 3: The Accessibility-Focused User

Name: Suresh Rao
Age: 60 years old
Occupation: Retired
Location: Chennai
Living Situation: Lives with family and manages household and personal expenses.

Physical & Digital Characteristics:

Suresh prefers larger text and clearly separated interface elements.

He may find complex financial terminology and dense information difficult to understand.

He benefits from simple visual summaries that communicate financial information without requiring detailed analysis.

Core Frustrations:

Small text can make financial information difficult to read.

Too much information displayed simultaneously can cause confusion.

Complex charts require additional effort to understand.

Unfamiliar financial terminology can make financial applications difficult to use.

Design Requirements:

Senior Mode with larger typography and expanded interaction areas.

High Contrast mode for improved readability.

Reduced Motion option.

Simplified Mode to reduce unnecessary information.

Clear financial summaries using simple and understandable language.

12. Questionnaire

NAME:PAWARYASWANTHRAO

REG NO : 23MIS0272

Basic User Information

1) **What is your age group?**

a. Below 18

b. 18–25

c. 26–40

d. 41–60

e. Above 60

2) **What is your current occupation?**

a. Student

b. Employee

c. Freelancer

d. Business owner

e. Retired

f. Other

3) **How frequently do you manage your personal finances?**

a. Daily

b. Several times a week

c. Weekly

d. Monthly

e. Rarely

4) **Which device do you mainly use for managing your finances?**

a. Smartphone

b. Laptop

c. Desktop

d. Tablet

5) **How comfortable are you with using financial applications?**

a. Very comfortable

b. Comfortable

c. Neutral

d. Uncomfortable

e. Very uncomfortable

User Experience Research.

6. How do you currently keep track of your income and expenses?

Answer: Users may use different methods such as financial applications, bank statements, notes, or manual tracking. MindWallet aims to centralize this information in one dashboard.

7. Do you currently use any personal finance or expense-tracking application?

Answer: Some users may use existing finance applications, but the project identifies that many existing applications focus mainly on transaction recording rather than decision support.

8. What do you like about the finance application or method you currently use?

Answer: Users generally benefit from being able to record transactions and view financial information, but MindWallet aims to make this information simpler and more accessible.

9. What problems do you face while tracking your expenses?

Answer: Major problems include complex interfaces, cluttered dashboards, difficult financial terminology, and the effort required to manually record expenses.

10. How difficult is it for you to understand your overall financial situation?

Answer: It can be difficult because existing systems often present dense charts, tables, and financial jargon. MindWallet addresses this using a Financial Health Score and simplified visual summaries.

11. Do financial applications contain too much information for you?

Answer: Yes. Many existing applications create high cognitive load by presenting too much information at once. MindWallet uses clean card-based layouts and progressive disclosure.

12. What type of financial information do you find difficult to understand?

Answer: Users may find financial analytics, complex charts, financial health indicators, and technical financial terminology difficult to understand.

13. **Do financial terms or jargon make finance applications difficult to use?**

Answer: Yes. Financial jargon can be confusing, especially for users with low financial literacy. MindWallet therefore uses plain and friendly language.

14. **How important is a simple and clean interface?**

Answer: Very important. A simple interface reduces cognitive load and makes financial management easier for beginners, elderly users, and users with disabilities.

15. **What is the one major problem you would like a personal finance application to solve?**

Answer: The main problem is to make personal financial management simple, understandable, accessible, and useful for making better spending decisions.

Discover Stage

16. **What is your biggest challenge when managing your money?**

Answer: The major challenges are tracking expenses, maintaining budgets, understanding financial health, and making informed purchasing decisions.

17. **How often do you forget to record an expense?**

Answer: Users may forget to record expenses because manual entry can be inconvenient. MindWallet addresses this through a Quick Add Expense feature.

18. **What types of expenses do you spend the most money on?**

Answer: MindWallet is designed to track multi-category expenses, allowing users to identify where their money is being spent.

19. **Do you usually create a monthly budget?**

Answer: Yes, budgeting is an important requirement of MindWallet**.** The application provides monthly budget tracking and budget planning.

20. **If you create a budget, how often do you follow it?**

Answer: Users may struggle to consistently follow budgets. MindWallet provides visual feedback, reminders, financial insights, and health indicators to encourage better financial habits.

21. **What usually causes you to exceed your budget?**

Answer: One major reason is unplanned or impulse spending. MindWallet addresses this using the Purchase Simulator.

22. **Do you track recurring payments or subscriptions?**

Answer: Subscription management is one of MindWallet's proposed features. Users can track subscriptions and receive reminders for recurring payments.

23. **Have you ever purchased something and later regretted it?**

Answer: Yes, impulse purchases can negatively affect budgets**.** MindWallet's Purchase Simulator is designed to help users evaluate a purchase before spending money.

Explore Stage

24. Would you find a Quick Add Expense feature useful?

Answer: Yes. It would make daily expense recording faster and more efficient.

25. Which information would you prefer to see immediately on the home dashboard?

Answer: The MindWallet dashboard should prioritize:

- Current balance
- Income
- Expenses
- Monthly budget
- Savings progress
- Financial Health Score
- Recent transactions
- Upcoming subscriptions

26. Would you prefer financial information to be displayed using charts, cards, numbers, or a combination?

Answer: A combination of cards, numbers, progress indicators, and simple charts would be most appropriate because MindWallet aims to reduce cognitive load.

27. Would a Financial Health Score help you understand your financial condition?

Answer: Yes. The Financial Health Score provides an intuitive summary of financial stability without requiring users to understand complicated financial terminology.

28. How useful would a Purchase Simulator be before making an expensive purchase?

Answer: Very useful. It can help users understand the potential impact of a purchase on their budget and financial health before spending.

29. What information would you want the Purchase Simulator to show?

Answer: It should show:

- Purchase amount
- Current available balance
- Remaining budget
- Effect on savings

- Effect on financial health
- Whether the purchase is affordable
- Guidance before making the purchase

30. Would reminders for upcoming bills and subscriptions be useful?

Answer: Yes. Reminders can help users avoid missing recurring payments and manage subscriptions effectively.

31. Would you prefer simple language instead of technical financial terminology?

Answer: Yes. MindWallet specifically uses plain language and friendly guidance to support users with low financial and digital literacy.

Listening Stage s User Research

32. Can you describe how you normally manage your money?

Answer: Users generally need to track income, record expenses, create budgets, monitor savings, manage subscriptions, and evaluate future purchases. MindWallet combines these activities into one centralized platform.

33. Tell us about the last time you had difficulty managing your budget.

Answer: A common difficulty is unexpected or impulse spending, which can cause users to exceed their planned budget. MindWallet addresses this with budgeting tools and the Purchase Simulator.

34. What frustrates you most when using financial applications?

Answer: The main frustrations are:

- Complex navigation
- Too much information
- Financial jargon
- Poor accessibility
- Small interface elements
- Lack of decision-support features

35. What would make you trust a personal finance application?

Answer: Users would need clear calculations, transparent feedback, simple explanations, confirmation dialogs, safe error handling, and reliable financial information.

36. Can you describe a financial decision where you wished you had more information before spending money?

Answer: Users may want to know how a large purchase will affect their remaining budget, savings, and overall financial health. This is the main purpose of the Purchase Simulator.

37. If you could change one thing about existing finance applications, what would it be?

Answer: Users would benefit from replacing complicated interfaces with a simple, accessible, human-centered interface that provides decision support instead of only recording

transactions.

User Research Steps Benefits

38. How important is it for you to see your financial information in one place?

Answer: Very important. MindWallet provides a centralized dashboard containing financial health, budgets, transactions, savings, and subscriptions.

39. Would personalized financial suggestions help you manage your money better?

Answer: Yes. Contextual financial guidance and smart insights can help users make better budgeting and purchasing decisions.

40. Would visual progress toward savings goals motivate you to save more?

Answer: Yes. MindWallet uses the Piggy Bank metaphor and visual progress tracking to make savings goals understandable and motivating.

41. How important are notifications for budgets, bills, and subscriptions?

Answer: Important. Notifications and reminders can help users maintain budgets and remember recurring financial obligations.

42. Would you be interested in testing a prototype of a new personal finance application?

Answer: Yes. Prototype testing is an important part of MindWallet's User-Centered Design process. User feedback will be used to evaluate and iterate the design.

User Personas

43. What is your primary financial goal?

Answer: The primary goals identified for MindWallet users include:

- Controlling expenses
- Saving money
- Maintaining a budget
- Reducing unnecessary spending
- Achieving savings goals
- Improving overall financial health

44. What motivates you to manage your finances?

Answer: Users are motivated by the desire to control spending, increase savings, avoid financial stress, achieve financial goals, and make confident purchasing decisions.

45. What is your biggest frustration with managing money?

Answer: The biggest frustration is understanding and organizing financial information while avoiding unnecessary spending.

46. How much time do you usually spend managing your finances each week?

Answer: MindWallet aims to reduce the time and effort required through Quick Add Expense, centralized dashboards, guided workflows, and simplified interfaces.

47. How important is accessibility when using a financial application?

Answer: Very important. Accessibility is a core requirement of MindWallet. The project targets **WCAG 2.2 AA** and considers elderly users and people with visual, motor, and cognitive impairments.

48. Which accessibility features would be useful to you?

Answer: MindWallet proposes:

- Larger font sizes
- High-contrast mode
- Dark mode
- Color-blind-friendly palettes
- Reduced motion
- Keyboard navigation
- Visible focus indicators
- Screen-reader compatibility
- Simplified Senior Mode

49. What type of financial user best describes you?

Answer: MindWallet is designed for beginner, intermediate, and advanced users, but it particularly focuses on users with low financial or digital literacy who need a simple interface.

50. If MindWallet were available today, what three features would you want most?

Answer: Based on the project objectives, the three most important features would be:

1. **Financial Health Score**
2. **Purchase Simulator**
3. **Budget and Expense Tracking**

13. Conclusion

MindWallet translates cognitive design principles into concrete interface and implementation decisions. Its financial features are organized around concepts that users can understand immediately, while actions are supported through clear navigation and structured controls. The system also provides continuous visual feedback so that users can see the consequences of their actions and evaluate their financial position.

A major strength of the implementation is the way it shifts part of the cognitive workload from the user to the application. Transactions, budgets, savings goals and subscriptions act as persistent external records, while analytics, health scoring and simulation perform processing that would otherwise require additional mental effort. The result is an interface intended to make personal financial management more understandable, less cognitively demanding and more decision-oriented.